

CHEM 3610 (Rowley) Midterm Exam 1 Equations and Constants

Gas Constant Values			
8.314	$\frac{J}{mole\ K}$	0.08314	$\frac{L\ bar}{mole\ K}$
0.08206	$\frac{L\ atm}{mole\ K}$	8.314	$\frac{m^3\ Pa}{mole\ K}$
Boltzmann Constant Values			
1.381×10^{-23}	$\frac{J}{K}$	0.6950	$\frac{cm^{-1}}{K}$

Conversions		
1 L atm	=	101.325 J
1 atm	=	1.01325 bar
1 atm	=	760 torr
1 atm	=	101,325 Pa

$$H = U + pV$$

$$dU = dq + dw$$

$$dw = -p_{external}dV$$

$$p = \frac{nRT}{V - nb} - a\frac{n^2}{V^2}$$

$$v_{rms} = \left(\frac{3RT}{M}\right)^{1/2}$$

$$\Delta U_V = C_V \Delta T$$

$$z = \sigma v_{rel} \mathcal{N}$$

$$\mathcal{N} = \frac{N}{V} = \frac{p}{k_B T}$$

$$\Delta H_{rxn}(T_2) = \Delta H_{rxn}(T_1) + \int_{T_1}^{T_2} \Delta C_p dT$$

$$\Delta H_{rxn}^\circ = \sum_{products} \nu_i \Delta H_{f,i}^\circ - \sum_{reactants} \nu_j \Delta H_{f,j}^\circ$$

$$p_i V_i^\gamma = p_f V_f^\gamma \quad \gamma = \frac{C_{p,m}}{C_{V,m}}$$

$$V_i T_i^c = V_f T_f^c \quad c = \frac{C_{V,m}}{R}$$

$$dU = \left(\frac{\partial U}{\partial V}\right)_T dV + \left(\frac{\partial U}{\partial T}\right)_V dT = \pi_T dV + C_V dT$$

$$dH = \left(\frac{\partial H}{\partial p}\right)_T dp + \left(\frac{\partial H}{\partial T}\right)_p dT = -\mu C_p dp + C_p dT$$

$$Z = \frac{pV}{nRT} = \left(1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \dots\right)$$

$$C_{V,m} = \frac{1}{2} R \cdot n_{D.o.F}$$

$$C_{p,m} = C_{V,m} + R$$

$$v_{mean} = \left(\frac{8RT}{\pi M}\right)^{1/2}$$

$$\Delta H_p = C_p \Delta T$$

$$pV = nRT$$

$$v_{rel} = \sqrt{2} v_{mean}$$

$$\lambda = \frac{v_{rel}}{z} = \frac{k_B T}{\sigma p}$$

$$\Delta H \approx \Delta U + \Delta n_{gas} RT$$

$$\pi_T = \left(\frac{\partial U}{\partial V}\right)_T$$

$$\alpha = \frac{1}{V} \left(\frac{\partial V}{\partial T}\right)_p$$

$$\kappa_T = -\frac{1}{V} \left(\frac{\partial V}{\partial p}\right)_T$$

[illegible]